

Tharindu Madushanka Samarakoon

Master's Student in Embedded Computing Systems RPTU

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Education

Technical University of Kaiserslautern-Landau, Germany

2024–Present

MSc. Embedded Computing Systems

- Current GPA - 1.202 (Semester 1–3)

University of Moratuwa, Sri Lanka CGPA: 4.01/4.20 (1st Class)

2017–2022

BSc. (Hons) Engineering – Electronic and Telecommunication Engineering

- Dean's List: Semester 1–8

Ananda College, Colombo-10, Sri Lanka

Grad: 2016

G.C.E. Advanced Level Examination (Mathematics, Physics, Chemistry, General English)
(country-wide university entrance examination taken by over 100,000 students annually)

- District Rank – 6
- Island Rank – 21
- 4As
- Z-Score – 2.7044

Experience

Integrated Automotive UK

February 2023–May 2026

FPGA Design Engineer (part-time working student)

Integrated Automotive is an automotive electronics company. I am involved in their pioneering research on analog signal analysis, contributing to the development of a dynamic signal analyzer and precision measurement tool.

- Developed SoC based hardware and firmware to:
 - Drive high-precision analog signals with a dynamic range of frequencies.
 - Capture and analyze the feedback signals with DFT and FFT units.
- Designed and implemented a dynamically configurable FFT unit on a custom Zynq-7020 FPGA board, including a downsampler, anti-aliasing filter, Hann window (developed from scratch), and the Xilinx FFT IP core. Developed an end-to-end pipeline from data capture and FFT analysis to result transfer via Ethernet (100 FFTs/s) for processing and via CAN bus (10 FFTs/s) for visualization.
- Working on unit-level and system-level verification.
 - Tools – Zynq7000, Vivado, Vitis, Altium, Kicad
 - Platforms – Petalinux and Kernal modules, FreeRTOS, Bare-metal
 - Languages – SystemVerilog, Verilog, C, Python
 - Protocols – AXI, CAN, SPI, I2C

SiFive, Inc. US

August 2022–September 2024

Verification Engineer (contract)

SiFive, Inc. is a fabless semiconductor company and provider of commercial RISC-V processor IP and silicon chips based on the RISC-V ISA.

- Worked on cache sub-system unit-level verification using UVM to verify cache coherency and consistency.
- Worked with Linux kernel, U-Boot and have emulated standard Linux tests with high-end prototyping devices.
- Automated the process of debugging using synthesizable assertions, designing an efficient debugging module architecture.
- Improved emulation test monitoring process by automating most of the manual tasks.
- Worked on developing synthesizable assertions to monitor newly designed **TileLink2** cache coherent bus protocol.
 - Tools – UVM, Synopsys tools (VCS, Verdi, ProtoCompiler, HAPS-100), Vivado, Jenkins
 - Languages – SystemVerilog, Chisel, TCL, Bash, Python, Google App Script

Zone24x7 is a strategy and implementation experts offering insights to global retail, manufacturing with measurable ROIs. Embedded Systems.

- **Research project** – Design a body area network to measure human mental situation and provide medical solutions.
- **Designed a 4-layer PCB** for a wearable device to acquire ECG, PPG, EDA signals from human body.
- Worked with ECG, PPG, EDA sensors. (AS7026GG, AFE4900)
- **Generated artificial ECG, PPG signals** (60 bpm) and **sine waves (0.1Hz – 15Hz)** using variable PWM signals and RC low pass filters for emulation purposes.
 - Tools – Altium, Multisim
 - Languages – C++

Projects

HeiChips26: ASIC DNA Sequence Alignment Accelerator [↗](#)

2026 August

- Designed a hardware-accelerated *Smith-Waterman* DNA sequence alignment engine using a systolic-array architecture for ASIC implementation.
- Implemented and evaluated the RTL-to-GDSII flow using open-source EDA tools, targeting the *IHP SG13CMOS5L 130 nm* process at *100MHz frequency* as part of the HeiChips26 tapeout.
 - Languages/Tools: SystemVerilog, Yosys, LibreLane, OpenROAD, KLayout

HeiChips26: Custom RISC-V Processor [↗](#)

2026 August

- Designed a custom *RV32* RISC-V processor core in SystemVerilog with a multi-cycle datapath and memory-mapped accelerator interface.
- Optimized the architecture for resource-constrained FPGA implementation, achieving a footprint of only **246 LUTs/FFs**.
 - Languages/Tools: SystemVerilog, RISC-V, Yosys, Vivado, Librelane

InvisiSpec: Making Speculative Execution Invisible in the Cache Hierarchy [↗](#)

2025–2026

- Implement *InvisiSpec* microarchitectural defense [MICRO 2018] to mitigate speculative execution side-channel attacks by preventing speculative loads from affecting the data cache hierarchy.
- Integration and evaluation on the **RISC-V BOOM** out-of-order processor core.
 - Languages: Chisel, Scala

RISCV32I Pipelined Processor + UVM testbench [↗](#)

2025

- Implemented 5-stage pipelined processor which supports all RV32I instructions.
- Developed a UVM-based verification environment capable of running the RISCV-tests standard test-suite on the processor (DUT).
- This enhanced processor and verification framework is now used as the class project for the "Architecture of Digital Systems I" module of RPTU.
 - Languages – Chisel, Scala, SystemVerilog

RISCV32 pipelined processor with Branch Target Buffer [↗](#)

2025

- Supports all R-type and ADDI instructions.
- Include a two-way set associative Branch Target Buffer with LRU replacement policy.
 - Languages – Chisel, Scala, SystemVerilog

RISCV32 processor optimization

2025

- Understanding software tool chain. (Linker Script, Startup code, RV32IM ISA, MakeFile etc.)
- Implement software based optimizations.
- Implement RV32-M ISA extension.
- Implement Application Specific RISC-V instructions.
- Implement Formal verification with SystemVerilog Properties and Assertions.
 - Tools – Vivado, Modelsim, riscv-gnu-toolchain
 - Languages – SystemVerilog, C

RISCV32 5-stage pipelined processor design [↗](#)

2022

- Supports 37 RV32I instructions including LHW, LB, LBU, LHU, SB, SH.
- Supports RV32M standard integer multiplication and division extension.
- Minimum number of stalls with forwarding and branch decision taken at decode stage.
- Memory controller design to handle memory operations with minimum delay.
- FPGA implementation
 - Tools – Vivado, Quartus Prime, Modelsim
 - Languages – SystemVerilog, Verilog

Serial system-bus architecture design [↗](#)

2021

- Serial system-bus protocol which can handle up to 2 internal masters and 3 internal slaves.
- Two way communication with external devices through UART protocol.
- FPGA implementation - Altera DE2-115 board
 - Tools – Quartus Prime, QuestaSim
 - Language – SystemVerilog

Multi-core processor design [↗](#)

2021

- Parameterized such that the number of cores can be changed.
- Communication between FPGA and PC through UART protocol.
- FPGA implementation - Altera DE2-115 board
 - Tools – Quartus Prime, QuestaSim
 - Language – SystemVerilog

Object detection for self driving - BSc. Final year project [↗](#)

2021-2022

- Real time object (traffic lights, lane lines etc.) detection, tracking and deploying on a vehicle.
- Creation of traffic light, lane line benchmark datasets.
- Main contribution - Design a camera system which consists of 2 vision grade cameras.
- Experience - TensorRT, ROS, PyTorch, Pylon, Python3, C++, SOLIDWORKS, FFMPEG

Resonance frequency meter design

2020

- Measure the resonance frequency of a metal piece during sanding by a CNC machine to find the end of the process. (Used to manufacture vehicle parking sensors)
- two-layer SMD PCB design
- Experience – STM32 MCUs, CubeMX, CubeIDE, Altium, C++

Generalized light field refocusing - BSc. Research project [↗](#)

2021-2022

- Real time, interactive sparse and dense light field refocusing with less computational complexity with convolutional neural networks.
- Experience - MATLAB, PyTorch, Python

Courses & Certifications

- Introduction to FPGA Design for Embedded Systems, University of Colorado, Coursera [↗](#)
- FPGA Softcore Processors and IP Acquisition, University of Colorado, Coursera [↗](#)
- Functional Programming Principles in Scala, EPFL, Coursera [↗](#)
- Neural Networks and Deep Learning, DeepLearning.AI, Coursera [↗](#)
- Structuring Machine Learning Projects, DeepLearning.AI, Coursera [↗](#)
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization, DeepLearning.AI, Coursera [↗](#)

Publications

- Harindu Jayarathne, **Tharindu Samarakoon**, Hasara Koralege, Asitha Divisekara, Ranga Rodrigo, Peshala Jayasekara – **DualCam: A Novel Benchmark Dataset for Fine-grained Real-time Traffic Light Detection**, 2023 International Conference on Machine Learning and Applications, Florida, USA (ICMLA) [↗](#)

Skills

- **HDLs:** SystemVerilog UVM Chisel Verilog
- **OS & Firmware:** Petalinux Kernal modules FreeRTOS Bare-metal
- **RTL & VLSI design software:** Xilinx Vivado, Vitis Synopsys VCS, Verdi, ProtoCompiler ModelSim, QuestaSim
- **Familiar prototyping devices:** Xilinx Virtex UltraScale+ FPGA VCU118 Synopsys HAPS-100
- **Programming Languages:** C++, C Bash, Python, Tcl Google App Script Arduino
- **Software:** STM32 CubeMX, CubeIDE Altium, Kicad, Orcad Multisim MATLAB SolidWorks
- **Platforms and Frameworks:** Jenkins Automation ROS PyTorch TensorRT Jetson AGX Xavier developer kit

Achievements

- Awarded the **Deutschlandstipendium (scholarship) 2025–26**
- **Grand prize winners of IEEE Signal Processing Cup 2021** at the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2021. (Team member of the winning team T-Cubed) [↗](#)
- **Bronze Medalist, International Physics Olympiad 2017**, Yogyakarta, Indonesia
- **Represented National Team, Asian Physics Olympiad 2017**, Yakutsk, Russia.
- **Gold Medal, Sri Lankan Physics Olympiad 2016**, Sri Lanka.
- **All-Island 16th (Global 326th out of 5015) in IEEEExtreme 13.0** (24-Hour Hackathon Competitive Programming Competition) held by IEEE.
- **All-Island 16th in MoraExtreme 2019** (An All-Island competitive programming competition held by the University of Moratuwa)
- **Finalist at SLRC 2019** (This competition stands out as the premier robotics challenge at the university level.)
- **Finalist at SLIIT Robofest Competition 2019** (An island-wide Micromouse competition)

Extra-curricular Activities

Co-director

2019 - 2020

- Community service avenue, Rotaract Club, University of Moratuwa.
- Expanded 'Hand in Hand' project to help adult cancer patient in National Cancer Hospital, Sri Lanka. (**Gold Award – “Most Outstanding Fundraiser 2019/20 – Rotaract District 3220 Sri Lanka & Maldives”**)

Co-chairperson

2018 - 2019

Community service project - 'Hand in Hand 2018/19', Rotaract Club of University of Moratuwa, Sri Lanka.

- **Raised funds up to Rs. 2,090,500** towards the National Cancer Hospital, Sri Lanka.
- **Silver Award – “Most Outstanding Fundraiser 2018/19 – Rotaract District 3220 Sri Lanka & Maldives”**

Conducting workshops

- Embedded Systems Workshop at University of Kelaniya, Sri Lanka.
Content - How to program a PIC MCU using MPLAB X IDE and simulate using Proteus software.
- Kick-Starter on Embedded System Design at University of Moratuwa, Sri Lanka.
Content - PCB design best practices

References

- Dr. Ranga Rodrigo
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